23. $Ba(OH)_2$ dissociates to form

Ba^{2+} and $2OH^-$ ion

$$\alpha = \frac{(i-1)}{(n-1)}$$

$$i = \alpha (n-1) + 1$$

$\therefore$ n = i = 3 (for $Ba(OH)_2$, $\alpha = 1$)

24. 10% $\frac{W}{W}$ aqueous NaOH solution means that 10 g of sodium hydroxide in 100g solution

$$\text{molality} = \frac{\text{no. of moles of solute}}{\text{weight of solvent (in kg)}}$$

$$= \frac{\left(\frac{10}{40}\right)}{0.1} = \frac{0.25}{0.1} = 2.5 \text{ M}$$

25. $\alpha = \frac{(1-i)n}{(n-1)}$ (or) $\frac{n(i-1)}{(1-n)}$

option (c) is correct

26. Elevation of boiling point is more in the case of Na_3PO_4 (no. of ions 4 ; 3 Na^+, PO_4^{3-})

27. $\Delta T_f = i \times K_f \times m$

$$i = \frac{\Delta T_f \times M_2 \times W_1}{K_f \times W_2 \times 1000}$$

$$= \frac{3.64 \times 142 \times 45}{1.86 \times 5 \times 1000}$$

$i = 2.5$

$K_f = 1.86$

$W_2 = 5g$

$W_1 = 45$ g

$\Delta T_f = 3.64$

$M_2 = 142$

28. Equimolal aqueous solution of KCl also shows 2° C depression in freezing point.

option (a) is correct.

29. i = 0.54

$$\alpha = \frac{(1-i)\,n}{(n-1)}$$

$$= \frac{(1-0.54)\,2}{(2-1)}$$

$$= 0.46 \times 2$$

$\alpha = 0.92$

30. a) both assertion and reason are correct and reason is the correct explanation of assertion.

40.

Given

Molarity = 12 M HCl

density of the solution = 1.2 g L^{-1}

In 12M HCl solution, there are 12 moles of HCl in 1 litre of the solution.

$$\text{molality} = \frac{\text{no.of moles of solute.}}{\text{mass of solvent (in kg)}}$$

calculate mass of water (solvent)

mass of 1 litre HCl solution

= density × volume

= 1.2 gmL^{-1} × 1000 mL

= 1200 g

mass of Hcl = no. of moles of HCl × molar mass of HCl

= 12 mol × 36.5g mol^{-1}

= 438 g

mass of water = mass of HCl solution – mass of HCl

mass of water = 1200 – 438 = 762 g

$$\text{molality} = \frac{12}{0.762}$$

$= 15.75$m

41. C = 0.25 M

T = 370.28 K

$(\pi)_{glucose}$ = CRT

$(\pi) = 0.25 mol L^{-1} \times 0.082 L\ atm\ K^{-1} mol^{-1} \times 370.28 K$

= 7.59 atm

42.

$$\text{molality} = \frac{\text{no. of moles of solute}}{\text{mass of solvent (in kg)}}$$

$$\text{no. of moles of glycine} = \frac{\text{mass of glycine}}{\text{molar mass of glycine}}$$

$$= \frac{7.5}{75} = 0.1$$

$$\text{molality} = \frac{0.1}{0.5 \text{ Kg}}$$

$= 0.2$m

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